INCH POUND

MIL-P-70786 (AR)
w/AMENDMENT 4
10 July 2008
SUPERSEDING
MIL-P-70786 (AR)
w/AMENDMENT 3
1 February 1996

MILITARY SPECIFICATION

PALLET, STANDARD METAL

Inactive for new design after 17 July 1997

This specification is approved for use by the US Army Armament Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture of standard metal pallets for use in unitizing ammunition (see 6.1).

1.2 Classification. The standard metal pallets shall be of the following types (see 6.2).

Type I	Pallet, Standard Size 44” x 40”, Sheet Metal
Type II	Pallet, Standard Size 45 1/2” x 35”, Sheet Metal
Type III	Pallet, Standard Size 48” x 40”, Sheet Metal
Type IV	Pallet, Standard Size 53” x 42”, Sheet Metal
Type V	Pallet, Special Size 56” x 28”, Sheet Metal
Type VI	Pallet, Special Size 70.75” x 29.31”, Sheet Metal
Type VII	Pallet, Special Size 78.50” x 29.31”, Sheet Metal

2. APPLICABLE DOCUMENTS

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806 5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 8140

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2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	Cleaning and Preparation of Ferrous and Coated Surfaces for Organic Protective Coatings

MILITARY

MIL-C-53072	Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-A-70625	Automated Acceptance Inspection Equipment for Non-Electronic Components Design, Testing and Approval of

STANDARDS

FEDERAL

FED-STD-101	Test Procedures for Packaging Materials

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1235	Single and Multi Level Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-l261	Welding Procedures for Constructional Steel
MIL-STD-1660	Design Criteria for Ammunition Unit Loads

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in affect on the date of the solicitation.

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DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

AC200000423	CARC Finishing Requirements for Metal and Pallet Adapters
AC200000431	Pallet, Standard Size 44” x 40”, Sheet Metal
AC200000433	Pallet, Standard Size 45 1/2” x 35”, Sheet Metal
AC200000435	Pallet, Standard Size 48” x 40”, Sheet Metal
AC200000437	Pallet, Standard Size 53” x 42”, Sheet Metal
AC200000413	Pallet, Special 56” x 28”, Sheet Metal
ACV00164	Pallet, Special 70.75” x 29.31”, Sheet Metal
ACV00165	Pallet, Special 78.50” x 29.31”, Sheet Metal
ACV00307	Post Support Plate

(Copies of specifications, standards, drawings and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernmental documents which is current on the date of the solicitation

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D4138	Measurement of Dry Film Thickness of Protective Coating System by Destructive Means

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

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3.2 Pallets. The pallet assemblies shall comply with all requirements specified on drawings AC200000431, AC200000433, AC200000435, AC200000437, AC200000413, ACV00164 and ACV00165 as applicable and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Stacking strength. The pallets shall withstand a 16,000 pound minimum (min.) stacking load.

3.4 Welding. The welding shall comply with the requirements of MIL-STD-1261, Class 1. The welding procedure and workmanship specimens shall be submitted prior to production in accordance with MIL-STD-1261.

3.5 Weld integrity. Workmanship specimens in accordance with MIL-STD-1261 shall be produced on a monthly basis. If first article procedures and specimens were completed satisfactory, further workmanship specimen submissions may be waived. This may be done prior to the first submission or following any monthly specimen submission. Failure of any monthly workmanship specimen shall require return of specimen submissions to the government.

3.5.1 DELETED

3.5.2 DELETED

3.6 Painted surface. The painted surface of the pallets shall comply with the requirements of Drawing AC200000423 CHEMICAL AGENT RESISTANT COATING (CARC) and applicable drawings.

3.7 Unit load tests. The unit loaded pallets shall conform to the requirements specified in MIL-STD-1660 (see 4.5.3).

3.8 First article inspection. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.9 Workmanship.

3.9.1 Pallets. The pallets shall be regular, smooth, free from wrinkles, cracks, rough spots, burrs, sharp edges and any other defect that might affect the serviceability, durability and safety of the pallets. Weld splatter is permissible on non-load bearing surfaces provided it is firmly attached.

3.9.2 Paint coating. The finished coat shall cover all required surfaces and be continuous. Paint runs are permissible provided they are no more than 3 inches in length, solid and free from cracks and blisters. Excess thickness is acceptable in areas where multiple passes of the paint gun are required to obtain total coverage. Paint chips which expose primer or bare metal are acceptable provided they are not more than 0.25 inches in diameter and no more than two such

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paint chips on a pallet. Scratches which expose primer or bare metal are permissible provided they are no more than 1.5 inches in length and no more than 10 inches total on a pallet. Bare metal on contact surfaces of bottom of skids and in areas inaccessible to paint is permissible.

3.9.3 Parts. All parts shall be free of chips, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance, of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of nine (9) painted pallet assemblies, ten (10) unpainted pallet assemblies and six (6) each of every subassembly and component.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First Article Inspection
CLASSIFICATION OF DEFECTS & TESTS

TITLE Assemblies and components		SHEET 1 OF 2		DRAWING NUMBER As Applicable NEXT HIGHER Assembly
EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Stiffener (Dwg AC200000427)				
a) Overall width	6		3.2	Gage
b) Overall length	6		3.2	Gage
c) Evidence of poor workmanship	6		3.9	Visual
d) Material thickness	3		3.2	Gage
Skid (Dwg. AC200000428)				
a) Overall width	6		3.2	Gage
b) Overall length	6		3.2	Gage
c) Evidence of poor workmanship	6		3.9	Visual
d) Material thickness	3		3.2	Gage
Post (Dwg. AC200000429)				
a) Height	6		3.2	Gage
b) Evidence of poor workmanship	6		3.9	Visual
c) Material thickness	3		3.2	Gage
Deck (Applicable drawing)				
a) Overall width	6		3.2	Gage
b) Overall length	6		3.2	Gage
c) Evidence of poor workmanship	6		3.9	Visual
d) Material thickness	3		3.2	Gage

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TABLE I. First Article Inspection
CLASSIFICATION OF DEFECTS & TESTS

TITLE Assemblies and components		SHEET 2 OF 2		DRAWING NUMBER As Applicable NEXT HIGHER Assembly
EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Unpainted pallet assemblies (Applicable drawings) Examination for defects	10		3.2	4.4.2.1
Pallet assemblies (Applicable drawings) a) Examination for defects b) Stacking strength c) Unit load test	 6 6 3		 3.2 3.3 3.7	 4.4.2.2 4.5.1 4.5.3
Post Support Plate (Dwg ACV00307) 1/ a) Overall width b) Overall length c) Evidence of poor workmanship d) Material thickness	 6 6 6 3		 3.2 3.2 3.9 3.2	 Gage Gage Visual Gage
Notes: 1/ Pertain to dwg. ACV00164 and ACV00165				

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation of MIL-A-48078.

4.4.2 Examination. Inspection for critical defects (and major defects when so specified) shall be 100 percent. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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PARAGRAPH 4.4.2.1	TITLE Unpainted pallet assembly		SHEET 1 OF 1		DRAWING NUMBER See Note 2/
					NEXT HIGHER Assembly
Classification	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100 %	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Mating surface of weld incorrect 1/		1.0	3.2	Gage
102	Location of adapter mounting hole(s) in deck		1.0	3.2	Gage
103	Location of adapter mounting hole(s) in skid		1.0	3.2	Gage
104	Height of pallet		1.0	3.2	Gage
105	Overall length		1.0	3.2	Gage
106	Overall width		1.0	3.2	Gage
107	DELETED		1.0	3.2	Gage
108	Visual inspection of weld, unsatisfactory 1/		1.0	3.2	Visual
109	Material certification 3/		1.0	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.5	3.9.1, 3.9.3	Visual

Notes:
1/ Pertain to load bearing welds. These are the welds which join the post to the skid and the post to the deck. For corner support posts of ACV00164 and ACV00165, load bearing welds are the welds which join the post to the post support plate, post support plate to the deck and the post to the skid.
2/ Dwg. AC200000431, AC200000435, AC200000437, AC200000413, ACV00164 or ACV00165 applies, as applicable
3/ A Certified Material Test Report (CMTR) shall be submitted for all chemical and physical requirements for compliance with the applicable drawing.

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PARAGRAPH 4.4.2.2	TITLE Pallet assembly		SHEET 1 OF 1		DRAWING NUMBER See Note 1/
					NEXT HIGHER Assembly
Classification	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Protective coating thickness (total thickness minus primer thickness) (Dwg. AC200000423 2/)		1.0	3.2	Gage
102	Marking missing, incorrect or illegible		1.5	3.2	Visual
103	Paint adhesion test	4	0-1	3.6	4.5.4
104	Primer coat thickness (Dwg. AC200000423 2)		1.0	3.2	Gage
Minor					
201	Protective finish missing, incomplete or incorrectly applied		1.5	3.9.2	Visual
202	Evidence of poor workmanship (other than protective finish)		4.5	3.9	Visual

Notes:
1/ Dwg. AC200000431, AC200000433, AC200000435, AC200000437, AC200000413, ACV00164 or ACV00165 applies, as applicable.
2/ The primer thickness may be determined from a test panel, when representative of the part for CARC topcoat. When destructive testing is done on samples (e.g. ASTM-D4138), these samples may be touched up in accordance with MIL-C-53072.

4.4.3 Testing.

4.4.3.1 Stacking strength. (See 3.3), Major defect - The pallets shall be selected in accordance with the appropriate sampling plan as follows:

Lot Size	Sample Size
1 - 200 pallets	2
201 - 500 pallets	5
501 - 1500 pallets	8
1501 - or more pallets	12

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The samples shall be tested in accordance with 4.5.1. If any sample fails to meet the requirement, the lot shall be rejected. After three (3) consecutive lots have been found acceptable, every third lot shall be tested thereafter. If any of these lots are rejected, testing must be returned to three (3) consecutive lots. After three consecutive lots pass, return to testing every third lot.

4.4.3.2 DELETED

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in Table I., 4.4.2.1, 4.4.2.2 and in 4.5. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. When Automated Acceptance Inspection Equipment is to be used, the provisions of MIL-A-70625 shall apply. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Stacking strength. The pallet shall be stacked with the required stacking load for a period of one (1) hour minimum. After that time, the pallet shall not have any evidence of breaking, bending, buckling or failure of any of the welds. Any pallet which fails to meet the requirement shall be classed defective.

4.5.2 DELETED

4.5.3 Unit load test. (First article inspection only (see 6.2)). Unit loads for the pallet and applicable adapter shall be fabricated in accordance with MIL-STD-1660 with the following exceptions. The gross weight of the ammunition load shall be 2,500 pounds minus 50 pounds, evenly distributed over the deck. Assembly of the unit load shall include the applicable pallet adapters and containers for the contract. The following test shall be performed on the pallets in accordance with MIL-STD-1660, in the sequence indicated (4.5.3.1 through 4.5.3.4).

4.5.3.1 Repetitive shock test. The acceptance/rejection criteria specified in section 7 of Method 5019.1 in FED-STD-101 is mandatory.

4.5.3.2 Edgewise drop (rotational) test. The acceptance/rejection criteria specified in section 7 of Method 5008.1 in FED-STD-101 is mandatory.

4.5.3.3 Incline impact test. The acceptance/rejection criteria specified in section 7 of Method 5023 in FED-STD-101 is mandatory.

4.5.3.4 Sling compatibility test. The acceptance/rejection criteria specified in section 5 of MIL-STD-l660 shall apply.

4.5.4 Paint adhesion test. The paint (primer and top coat) adhesion shall be tested in accordance with TT-C-490.

5. PACKAGING

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5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. This metal pallet is intended for use with ammunition containers, projectiles and cartridges.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Type of pallet (see 1.2).

c. Applicable stock number.

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2).

e. First article sample requirements. Contractual provisions may require that the unit load test (as per MIL-STD1660) will be performed at the U.S. Army Defense Ammunitions Center and School (USADACS), SMCAC-DEV, Savanna, IL 61074-9639.

f. Packaging requirements, if other than specified in Section 5.

g. Serialization requirements, if applicable.

h. Certificate of conformance for each lot of product shipped.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATN: AMSTA-AR-QAT-P, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the contracts Data Requirement List, DD Form 1423 in the contract.

6.4 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSTA-AR-QAT-P, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the accuracy and precision of

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the method, test data demonstrating the accuracy and precision, and drawings of any special equipment required.

6.5 Subject term (key word) listing.
Ammunition
Cartridge
Container
Projectiles

6.6 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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APPENDIX

QUALITY ASSURANCE PROVISIONS FOR REFURBISHMENT

10. SCOPE:

10.1 Scope. This appendix provides the Quality Assurance provisions for the refurbishment of metal pallets.

20. APPLICABLE DOCUMENTS:

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-171	-Finishing of Metal and Wood Surfaces

30. REFURBISHMENT:

30.1 Cleaning. Loose paint, dirt, grease and rust shall be removed.

30.2 Refinish. The parts shall be finished in accordance with the applicable drawings.

40. REQUIREMENTS:

40.1 Welds. Welds shall be in accordance with good commercial practices.

40.2 Stackabilitv. The pallet shall not be deformed to the extent that it effects safely stacking of pallets in storage.

40.3 Workmanship. See paragraph 3.8 unless otherwise specified below:

a. Deformation or damage that does not effect stacking, slinging or assembly of the pallet is allowed.

b. Damage or deformation to the pallet legs that does not allow a four way entry or prevents the pallet from resting evenly on a level surface is not allowed.

50. INSPECTION:

50.1 Examination. Unless otherwise specified examinations for major and minor characteristics shall be performed in accordance with MIL-STD-l05, inspection Level II.

50.1.1 Metal Pallet (see drawings listed in 2.1.2 as applicable).

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CLASS	CHARACTERISTICS	AQL or 100%	INSPECTION METHOD
Critical	None Defined		
Major			
101	Welds missing or broken	1.0%	40.1/Visual
102	Stacking surface deformed	1.0%	40.2/Visual
Minor			
201	Marking missing, incorrect or illegible	1.0%	3.2/Visual
202	Protective finish damaged	1.0%	3.2/Visual
203	Poor workmanship	1.0%	40.3/Visual

Custodian
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